

GHOUSE MOHIUDDIN MOHAMMED

Data Engineer | AWS | ETL/ELT | Redshift/PostgreSQL | Python/SQL

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PROFESSIONAL SUMMARY

Data engineer with **8 months of production experience** building cloud data platforms end to end. At Apex Mobile Media, solo-built the company data platform: a **400M+ row** Amazon Redshift warehouse fed by **12+** partner pipelines, and a full-stack analytics application on AWS used daily by **about 60** people. Cut client reporting from **days to seconds**, nightly data migration by **99%**, and month-end finance close by **80%+**. Strengths across **ETL/ELT design**, data modeling and warehousing, **query performance engineering**, and containerized cloud deployment. Graduating December 2026 (MS Information Systems, GPA 3.84), seeking a full-time data engineering role.

WORK EXPERIENCE

Full Stack Developer Co-op

Jan 2026 – Aug 2026

Apex Mobile Media

Toronto, ON

- Cut client reporting from **3–4 days** of manual data pulls across every partner source to **seconds** by building the ingestion platform connecting **12+** partner sources (REST APIs, S3, email feeds) into a **400M+ row** Amazon Redshift warehouse.
- Reduced nightly data migration by **99%** (benchmarked at **about 24 hrs**, redesigned to **under 15 min**) by architecting a **hot/cold two-tier system**: monthly Parquet partitions in S3 with hash-fingerprint change detection and atomic partition swaps, so only changed data is re-pulled.
- Cut month-end finance billing preparation by **over 80%** (days → hours) by automating carry-forward pipelines, while preserving controller review for judgment-based adjustments.
- Delivered **100–1000x query speedups** (campaign pacing **411s** → **28ms**) through materialized-view rollups, targeted indexing, and connection-pool tuning.
- Shipped the analytics platform end to end (**FastAPI, React/TypeScript on AWS**) serving **about 60** daily users, with JWT authentication, **10-role RBAC**, and **30+** modules including automated Excel/PowerPoint report generation.
- Unlocked analytics the business previously had no access to: company-wide trend visibility for leadership, growth forecasting, projected-vs-actual sales tracking, and product mix per account executive.
- Ran **34M+ row** historical backfills with fault tolerance and rate-limit handling; led **pre-launch security audits** (authN/Z, SQL injection, secrets, dependency CVEs).
- **Skills**: Python, SQL, FastAPI, React, TypeScript, Amazon Redshift, PostgreSQL, DuckDB, Parquet, Docker, AWS (EC2, RDS, ALB, S3, ECR, CloudWatch, IAM), NetSuite ODBC, systemd, Linux, Jira, Confluence.

Software Engineer, Volunteer

Northeastern University

Toronto, ON

- Developed and deployed a menu-driven **Flask chatbot on Azure** for the Toronto campus at **\$0** infrastructure cost, collaborating with stakeholders on requirements and conversational flows meeting accessibility standards.
- **Skills**: Python, Flask, Microsoft Azure, REST API integration, cost-optimized cloud deployment, stakeholder collaboration.

TECHNICAL PROJECTS

Flight Refund RAG Assistant

- Building a **retrieval-augmented generation** system answering flight-cancellation refund questions grounded in airline policy and air-passenger regulatory documents, with **source-cited answers** to prevent hallucinated policy claims.
- Engineering the production envelope: **Docker**, FastAPI service layer, **CI**, and documented architecture with design trade-offs.

TECHNICAL SKILLS

Languages & Frameworks: Python (production-grade), SQL, Java, C, C++, TypeScript/JavaScript, React, Node.js

Data Engineering: ETL/ELT pipeline design, data modeling & warehousing, **change data capture** (hash-based), **incremental replication**, **hot/cold data tiering**, partitioning strategy, idempotent ingestion, retry/backoff, historical backfill, late-arriving data handling

Data Quality: Reconciliation & validation frameworks, anomaly detection & correction, audit logging, atomic partition swaps (zero-downtime data deployment)

Data Stores & Performance: Amazon Redshift (MPP), PostgreSQL, DuckDB, Parquet/columnar storage; **query optimization**, materialized-view rollups, indexing strategy, bulk-load patterns (UNLOAD/COPY)

AWS & Cloud: EC2, RDS, S3, ALB, ECR, CloudWatch, IAM, ACM; Docker, Linux, systemd, CI/CD, job scheduling & orchestration, observability & alerting

Application & API: FastAPI, React, SQLAlchemy, REST API design, JWT authentication & role-based access control (RBAC)

Integrations & Security: NetSuite (SuiteAnalytics ODBC), Gmail/Drive APIs, partner REST APIs; IAM least-privilege, secrets management, CVE remediation, container hardening

Collaboration & Tooling: Agile ceremonies, Jira, Confluence, architecture diagrams & design specs, benchmarking, AI-assisted development (Claude Code)

EDUCATION

Master of Science: Information Systems (College of Engineering)

Sep 2024 – Dec 2026

Northeastern University, Toronto, ON

GPA: 3.84 / 4.0

- Coursework: Software Engineering, Data Science Engineering Methods, Program Structure & Algorithms, Database Design

Bachelor of Technology: Information Technology

Jul 2019 – Jul 2023

Manipal Institute of Technology, Manipal, India